ASSESSMENT TOOL - 1

SIMATS ENGINEERING

COURSE : ARTIFICIAL INTELLIGENCE "C"

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ASSESSMENT 1

Water Jug problem

Let :

Jug A = 4 gallons

Jug B = 3 gallons

Steps:

1. Fill the 3-gallon jug completely (A = 0, B = 3)
2. Pour the water from the 3-gallon jug into the 4-gallon jug. (A = 3, B = 0)
3. Fill the 3-gallon jug again. (A = 3, B = 3)
4. Pour water from the 3-gallon jug until the 4-gallon jug is full (A = 4, B = 2)
5. Empty the 4-gallon jug onto the ground (A = 0, B = 2)
6. Pour the remaining 2 gallon jug contains exactly 2 gallons of water.

Final Answer:

The 4-gallon jug contains exactly 2 gallons of water.

2 Mars Rover as an Intelligent Agent

A mars Rover is a robot that explores the surface of mars. It collects samples, takes pictures, and sends information to Earth.

i. What are the percepts of this agent?

- Images of the martian surface
- Temp and weather conditions
- Rocks, soil, and other objects.
- Distance from obstacles.
- Chemical composition of samples.

ii. Characterize the operating environment.

The mars Rover operates in following

- Partially observable:
- stochastic
- sequential
- static
- continuous
- single-agent.

iii. What actions can the agent take?

1. move forward, backward, turn.
2. Take photo of the surface
3. collect rock and soil samples
4. analyse samples using scientific instrument.

5. Avoid obstacle

6. Send collected data to Earth

7. Recharge or conserve battery power when necessary.

iv. How can we evaluate the performance of the agent?

The rover's performance can be evaluated by:

* The amount of useful scientific data collected.
* The number of sample successfully analysed.
* The distance explored.
* The accuracy of observation
* The ability to avoid obstacles.
* Safe completion of its motion.

v. What is the most suitable agent architecture? Why?

A model-based agent is suitable for the Mars Rover because it maintains information about its surroundings and selects actions to achieve specific goals.

This architecture helps the rover make decisions, plan its movements, and complete its scientific mission safely.

3 Eight Queens problem

The eight queens problem is a search problem in AI. The goal is to place eight queens on a chessboard so that no two queens share the same row, column, or diagonal.

Problem formulation

1. Initial state

An empty 8 x 8 chessboard

2. State

3. Action

transition model.

Goal test

path cost

Search Strategy

The problem can be solved using Backtracking search.

Example solution

One valid arrangement, where each number represents the row containing the queen in the corresponding column is

[1, 5, 8, 6, 3, 7, 2, 4]

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OLA

Cab Booking using a Problem Solving Agent.

Type of Agent : Goal - Based Problem - Solving Agent.

A goal based problem - solving agent is suitable because the customer has a specific goal to reach destination.

Pseudocode

```
BEGIN
INPUT pickup_location
INPUT destination
INPUT preferred_cab_type
IF pickup_location is valid AND destination is valid THEN
    FIND available cabs
    IF preferred_cab_type is available THEN
        SELECT preferred_cab_types
    ELSE
        DISPLY other available cab types
        SELECT a suitable cab
    ENDIF
    CALCULATE estimated fare
    ESTIMATE travel time
    DISPLAY fare and travel time
    IF customer confirms booking THEN
        Book the cab
        DISPLAY driver and cab details
```

TRACK the cab

ELSE

CANCEL the booking

END IF

ELSE

DISPLAY "Invalid location"

END IF

END

Conclusion:

A goal-based problem-solving agent helps the customer choose a suitable cab and reach the destination.

5 Uniform cost search (UCS) Algorithm

Uniform cost search (UCS) is an uninformed search algorithm that finds the path with the lowest total cost from the destination warehouse G1 using uninformed cost search.

Pseudocode:

```
BEGIN
Insert s into the priority queue with cost 0
WHILE priority queue is not empty DO
    Remove the node with the lowest cost
    IF node is G1 THEN
        RETURN path and total cost
    END IF
    FOR each neighbour of the node DO
        calculate new path cost
        IF the new path is cheaper Then
```

```
        update the path cost
        Insert neighbour into the priority queue
    END IF
  ENDFOR
ENDWHILE
RETURN "Path not found"
END
```

Conclusion:

Uniform cost search finds the least-cost path by always expanding the node with the lowest cumulative cost.